Here's a concise one-page project summary with a focused 3-month work plan.

B.Tech Project Summary

Title

Development of a Probabilistic Motor Subtype Progression Model for Parkinson's Disease Using the PPMI Dataset

Background

Parkinson's disease (PD) is clinically classified into three motor subtypes: **Tremor Dominant (TD)**, **Postural Instability and Gait Difficulty (PIGD)**, and **Indeterminate**. These subtypes are calculated from a small set of MDS-UPDRS tremor and gait/postural instability items using predefined score ratios.

Although widely used in clinical practice, this classification represents only the patient's current motor state. Longitudinal studies have shown that patients can transition between these subtypes as the disease progresses. However, the current classification does not indicate how likely a patient is to remain in the same subtype or progress to another.

Project Objective

The objective of this project is to develop a probabilistic progression model using longitudinal data from the Parkinson's Progression Markers Initiative (PPMI).

The model will use the existing TD/PIGD classification framework and learn transition patterns from longitudinal follow-up data. Rather than proposing a new classification system, the goal is to estimate the probability that a patient will:

- remain in the current motor subtype, or
- transition to another subtype over time.

The initial model will use only the clinical variables involved in the current TD/PIGD classification and their longitudinal evolution.

Hypothesis

Patients with the same current TD/PIGD classification do not necessarily have the same future disease trajectory. Longitudinal modeling of subtype transitions can provide individualized probabilities of progression that complement the existing clinical classification.

Work Plan (First 3 Months)

Weeks 1 & 2 : Understanding the Dataset

- Read about the TD/PIGD classification method.
- Identify the MDS-UPDRS (Movement Disorder Society–Unified Parkinson's Disease Rating Scale) items used in calculating the TD/PIGD ratio.
- Explore the longitudinal structure of the PPMI dataset.
- Generate TD, PIGD, and Indeterminate labels for each patient visit.

Deliverable: Clean longitudinal dataset with subtype labels for every visit.

Weeks 3-4 : Longitudinal Analysis

- Track subtype changes for each patient across visits.
- Compute transition frequencies (e.g., TD → TD, TD → Indeterminate, TD → PIGD).
- Estimate transition probabilities over different follow-up intervals.
- Visualize common progression pathways.

Deliverable: Transition matrices and visualizations of subtype progression.

Weeks 5-7 : Baseline Probabilistic Model

- Develop a simple probabilistic model (e.g., Markov model or transition probability model) using the observed longitudinal transitions.
- Estimate, for each current subtype, the probability of remaining stable or transitioning at the next follow-up.
- Validate the model using held-out patients.

Deliverable: Initial progression model providing probabilities of subtype transition.

Expected Outcome

At the end of the first three months, the project should produce a working proof-of-concept model that augments the current TD/PIGD classification with probabilities of future subtype progression. This will serve as the foundation for more advanced predictive models during the remaining months of the project.

I deliberately kept the first phase focused on **understanding and modeling the subtype transitions**, rather than exploring all clinical variables. This gives the students a clear, achievable goal. The second half of the project can then build on this foundation by introducing more sophisticated probabilistic models (e.g., Hidden Markov Models, survival analysis, or Bayesian approaches) and evaluating whether additional baseline features improve prediction.

Further we can see if we can make the model powerful enough to predict the timeframe within which the subject may transition to next state ie from TD to PIGD...Which seems to be the most important transition that clinicians may want to be warned of.

Proposed Modeling Framework Summary

The proposed work aims to develop a **hierarchical Bayesian longitudinal model** that extends the existing clinical classification of Parkinson's disease motor subtypes (Tremor Dominant (TD), Indeterminate, and Postural Instability and Gait Difficulty (PIGD)) by providing individualized probabilistic predictions of disease progression.

The key idea is to retain the current clinically validated TD/PIGD ratio rather than introducing a new disease score. At each visit, the observed ratio is treated as a noisy measurement of a patient's underlying (latent) motor phenotype. The model assumes that this latent phenotype evolves over time according to a population-level progression process while allowing each patient to have their own progression trajectory through subject-specific random effects. In this way, the model learns from the entire cohort while simultaneously accounting for inter-subject variability and irregular longitudinal sampling.

Once the model is trained, the posterior distribution of a patient's latent motor phenotype can be propagated forward in time to estimate the probability of remaining in the current subtype or transitioning to another subtype at future follow-up visits. Thus, instead of assigning only a deterministic subtype, the model provides clinically interpretable outputs such as:

- Probability of remaining TD at the next annual visit.
- Probability of transitioning to the Indeterminate state.
- Probability of progressing to PIGD.

The framework therefore converts the current rule-based classification into a probabilistic disease progression model while remaining fully compatible with existing clinical practice.

Expected Outcomes: The primary outcome will be a validated probabilistic model that estimates future motor subtype progression from longitudinal clinical data. In addition to transition probabilities, the model will estimate patient-specific progression trajectories and quantify uncertainty in these predictions. Model performance can be evaluated by comparing predicted and observed subtype transitions on held-out patients and by assessing calibration of the predicted probabilities.

Future Extensions: The proposed framework naturally supports incorporating additional baseline information, such as demographics, imaging, biomarkers, genetics, or digital phenotyping, as predictors of progression. Beyond Parkinson's disease, the same hierarchical Bayesian framework could be generalized to model longitudinal progression in other neurological disorders in which clinical staging relies on repeated measurements with inherent uncertainty.

Implementation of a **hierarchical Bayesian motor phenotype progression model**

Model Development Roadmap

The implementation will be carried out in successive stages, beginning with a simple probabilistic model and progressively incorporating the hierarchical Bayesian framework.

Stage 1: Data Preparation

The longitudinal clinical data from the PPMI dataset will be organized in a subject-wise format. For each visit, the TD/PIGD ratio will be computed using the standard MDS-UPDRS items currently employed in clinical classification. Each record will therefore contain:

- Subject ID
- Visit time
- TD/PIGD ratio
- Clinical subtype (TD, Indeterminate, or PIGD)

This dataset will serve as the input for all subsequent analyses.

Stage 2: Exploratory Longitudinal Analysis

The longitudinal behaviour of the TD/PIGD ratio will be analysed to understand:

- Typical progression trajectories.
- Distribution of subtype transitions.

- Variability within and across subjects.
- Effect of unequal follow-up duration and missing visits.

This step will guide the selection of suitable prior distributions and model assumptions.

Stage 3: Hierarchical Bayesian Model

A hierarchical Bayesian model will then be constructed with two levels:

Population level: estimates the average disease progression trajectory across all patients.

Subject level: estimates patient-specific deviations from the population trajectory using random effects, allowing each patient to progress at a different rate while borrowing statistical strength from the entire cohort.

The observed TD/PIGD ratio at each visit will be modelled as a noisy observation of an underlying latent motor phenotype that evolves over time.

Stage 4: Posterior Inference

Bayesian inference will be used to estimate the posterior distribution of the latent motor phenotype for every patient at every visit. Posterior predictive inference will then be used to estimate the latent phenotype at future visits.

Stage 5: Probabilistic Subtype Prediction

The predicted posterior distribution of the future TD/PIGD ratio will be converted into clinically meaningful probabilities by calculating the probability that the predicted ratio lies within the existing TD, Indeterminate, or PIGD classification ranges.

The final output for each patient will be of the form:

- Probability of remaining in the current subtype.
- Probability of transition to another subtype at the next follow-up.
- Credible intervals reflecting prediction uncertainty.

Stage 6: Model Validation

The model will be evaluated using longitudinal cross-validation by withholding future visits during training and comparing predicted subtype probabilities with the observed clinical outcomes. Model calibration, predictive accuracy, and uncertainty estimates will be assessed to determine clinical reliability.

Stage 7: Model output visualization

So the output to the clinician could be:

Time	TD	Indeterminate	PIGD
Current	92%	8%	0%
1 year	78%	18%	4%
2 years	56%	28%	16%
3 years	31%	37%	32%
5 years	12%	23%	65%

Overall Modelling Strategy

The proposed framework uses a **Hierarchical Bayesian Longitudinal Model** to estimate disease progression from repeated clinical observations.

Rather than treating the observed TD/PIGD ratio as the patient's true motor state, the model assumes that each observed ratio is a noisy measurement of an underlying (latent) motor phenotype.

Conceptually,

Observed TD/PIGD Ratio = True TD/PIGD Ratio + Measurement Noise

The measurement noise accounts for sources of variability such as clinical scoring differences, medication effects, day-to-day symptom fluctuations, and other unexplained variability.

Disease Progression Model

The underlying motor phenotype is assumed to evolve continuously over time. Instead of modelling each patient independently, the model learns progression at two levels simultaneously.

Population Level

The model first estimates the average disease progression trajectory from the entire cohort.

This represents the expected evolution of motor phenotype for a typical Parkinson's disease patient.

Subject Level

Each patient is then allowed to deviate from this average trajectory.

Conceptually,

Bayesian hierarchical state-space (dynamic linear) model

True Ratio(t+1) = True Ratio(t) + Population Progression + Patient-specific Progression + Random Disease Noise

Or

Bayesian hierarchical growth model **(I think this is better- added a detailed implementation later)**

Observation model

Growth model

This hierarchical structure enables **partial pooling**, in which patients with fewer longitudinal visits benefit from information learned across the full cohort while still retaining their individual progression characteristics.

Bayesian Inference

Using all longitudinal observations, Bayesian inference estimates the posterior distribution of the latent motor phenotype for every patient.

Instead of estimating a single value,

True Ratio = 1.08

the model estimates

True Ratio ~ Probability Distribution

The width of this distribution naturally reflects confidence in the estimate.

Patients with stable longitudinal measurements will have smaller uncertainty, whereas patients with highly variable clinical measurements will have larger uncertainty.

Predicting Future Progression

Once the posterior distribution of the latent motor phenotype has been estimated, the disease progression model is used to predict future motor states.

Conceptually,

Current Posterior Distribution

↓

Progression Model

↓

Predicted Distribution after 1 year, 2, 3 or 4 years

Predicted Distribution after 2 years

↓

Unlike conventional classifiers, the model predicts an entire probability distribution rather than a single future subtype.

Clinical Interpretation

The predicted posterior distribution is translated directly into the existing clinical classification using the established TD/PIGD thresholds.

For example,

- **P(TD)** = Probability that the predicted TD/PIGD ratio is greater than 1.15
- **P(Indeterminate)** = Probability that the predicted ratio lies between 0.90 and 1.15
- **P(PIGD)** = Probability that the predicted ratio is less than 0.90

The clinician therefore receives outputs such as

Current Subtype : TD

Probability of remaining TD next year 82%

Probability of becoming Indeterminate 15%

Probability of becoming PIGD 3%

The existing clinical definitions remain unchanged; the proposed framework simply attaches individualized probabilities to them.

Interpretation of Model Parameters

One of the major advantages of the hierarchical Bayesian framework is that every model parameter has a direct interpretation.

Population progression

Represents the average rate of motor progression across the entire Parkinson's disease cohort.

Patient-specific progression

Represents how much faster or slower an individual progresses relative to the average patient.

Measurement variance

Represents uncertainty arising from clinical assessment and observation noise.

Posterior variance

Represents confidence in the estimated motor phenotype for an individual patient.

These parameters are intuitive and can be communicated directly to clinicians.

Expected Outputs

For each patient, the model will provide

- Current estimated motor phenotype.
- Confidence associated with the current estimate.
- Probability of remaining in the current subtype.
- Probability of transition to the other motor subtypes over future follow-up visits.
- Individual disease progression trajectory relative to the population trajectory.

Future Extensions

Once the longitudinal framework has been validated using the TD/PIGD ratio alone, additional baseline information such as demographics, imaging, biomarkers, genetics, cognitive measures, or digital phenotyping can be incorporated as predictors of patient-specific progression.

The same framework can also be extended to other longitudinal neurological disorders in which disease staging is based on repeated clinical assessments, which are inherently uncertain.

Scientific Contribution

The proposed work introduces a probabilistic extension to the current TD/PIGD classification by combining hierarchical Bayesian modeling with longitudinal disease progression analysis. Rather than redefining motor subtypes, it preserves the existing clinically validated framework while providing individualized estimates of future progression and associated uncertainty. This transforms the current rule-based classification into a clinically interpretable decision-support tool that can

assist neurologists in prognosis, patient monitoring, and personalized disease management.

Hierarchical Bayesian Growth Model

1. Observation Model

The observed TD/PIGD ratio is assumed to be a noisy measurement of the patient's underlying (latent) TD/PIGD ratio.

$$R_{it}^{obs} = R_{it}^{true} + \epsilon_{it}^{obs}$$

where

- R_{it}^{obs} : observed TD/PIGD ratio for subject i at visit t
- R_{it}^{true} : latent (true) TD/PIGD ratio
- $\epsilon_{it}^{obs} \sim \mathcal{N}(0, \sigma_{obs}^2)$

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2. Hierarchical Growth Model

The latent TD/PIGD ratio is modeled as a combination of a population trajectory and subject-specific deviations.

$$R_{it}^{true} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t + \epsilon_{it}^{proc}$$

where

- β_0 : population baseline TD/PIGD ratio
- β_1 : average population progression rate
- b_{0i} : subject-specific baseline deviation (random intercept)
- b_{1i} : subject-specific progression rate (random slope)
- $\epsilon_{it}^{proc} \sim \mathcal{N}(0, \sigma_{proc}^2)$

3. Random Effects

The patient-specific effects are assumed to arise from the population.

$$b_{0i} \sim \mathcal{N}(0, \tau_0^2)$$

$$b_{1i} \sim \mathcal{N}(0, \tau_1^2)$$

This allows every patient to have an individual disease trajectory while borrowing statistical strength from the entire cohort (partial pooling).

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4. Bayesian Inference

Using all longitudinal observations, estimate

$$P(R^{true}, \beta, b, \sigma, \tau \mid R^{obs})$$

to obtain the posterior distribution of each patient's latent TD/PIGD ratio and progression trajectory.

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5. Future Prediction

The posterior predictive distribution at a future visit is computed as

$$P\left(R_{i,t+\Delta}^{true} \mid R_{i,1:t}^{obs}\right)$$

which provides the complete probability distribution of the patient's future TD/PIGD ratio.

6. Clinical Translation

The posterior predictive distribution is mapped onto the existing clinical subtype definitions:

$$P(TD) = P(R^{true} > 1.15)$$

$$P(Indeterminate) = P(0.90 \leq R^{true} \leq 1.15)$$

$$P(PIGD) = P(R^{true} < 0.90)$$

The clinician therefore receives predictions such as:

- **Probability of remaining TD:** 82%
- **Probability of transitioning to Indeterminate:** 15%
- **Probability of progressing to PIGD:** 3%

One suggestion for the manuscript

I would actually simplify the notation even further by **dropping the "true" superscript** in the equations. Since the observation model already distinguishes the observed ratio from the latent ratio, the rest of the model can simply use R_{it} to denote the latent quantity.

That gives a very clean generative model:

$$\boxed{R_{it}^{obs} = R_{it} + \epsilon_{it}^{obs}}$$

$$\boxed{R_{it} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})t + \epsilon_{it}^{proc}}$$

This is concise, mathematically complete, and immediately recognizable to anyone familiar with Bayesian hierarchical mixed-effects models. I think this is the version I'd put in the proposal or the Methods section of the paper.

Extension or second half of the paper

I actually think this is where your work can move beyond "another hierarchical Bayesian model" into something that is methodologically interesting.

The key idea is that **progression** and **stability of progression** are two different patient characteristics.

Currently, your model assumes everyone shares the same residual variance

This means the model assumes that every patient fluctuates around their trajectory by roughly the same amount.

I don't think that's biologically reasonable.

Instead

Suppose every patient has their own variability, which becomes another latent parameter.

Now every patient has

- baseline ratio
- progression rate
- variability around progression

So every patient is characterized by

Patient A

Baseline ratio 2.8, Progression slope -0.12/year, Variability 0.08

Patient B

Baseline ratio 2.8, Progression slope -0.12/year, Variability 0.45

Interpretation: Exactly the same progression but very different stability.

Now your model becomes a hierarchical prior model

Now your model becomes

Observation

$$R_{it}^{obs} = R_{it} + \epsilon_{it}^{obs}$$

Growth

$$R_{it} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})t + \epsilon_{it}$$

Residual

$$\epsilon_{it} \sim N(0, \sigma_i^2)$$

Hierarchical variance

$$\log(\sigma_i) \sim N(\mu_\sigma, \tau_\sigma^2)$$

Now every patient has

three latent characteristics

instead of two.

Why is this exciting?

Suppose

Patient A has very low variability. Future prediction

TD > TD> TD > PIGD

Very confident.

Patient B same slope but huge variability. Prediction

TD > TD or Intermediate > Intermediate > PIGD

Much more uncertain.

Now your probabilities naturally widen.

Here's the really novel idea

Instead of treating uncertainty as merely a statistical quantity, treat it as a **patient phenotype**.

That is,

patients are characterized by

1. Baseline disease severity
2. Disease progression rate
3. Disease stability (or variability)

The third quantity is estimated directly from the longitudinal data.

Then the manuscript becomes stronger

The first half of the paper develops the hierarchical Bayesian progression model.

The second half asks: **Does longitudinal motor variability predict future subtype progression?**

For example,

fit

$$P(\text{PIGD in 3 years}) = f(\text{baseline ratio}, \text{slope}, \sigma_i)$$

Now you test

Is

$$\sigma_i$$

an independent predictor?

If yes,

that's a genuinely new result.

I would actually change the conceptual framework slightly

Instead of saying the model estimates **the TD/PIGD ratio**, I would say it estimates **three latent patient characteristics**:

Latent parameter	Clinical meaning
Baseline ratio (b_{0i})	Current motor phenotype
Progression slope (b_{1i})	Speed of disease progression
Longitudinal variability (σ_i)	Stability of motor phenotype over time

The clinical outputs (probability of remaining TD, becoming Indeterminate, or progressing to PIGD) are then derived from **all three parameters**, not just the mean trajectory.

I think this is the point where your work starts to have a distinct identity. Most progression models estimate where the patient is headed. Your framework would also estimate **how predictable that journey is**. That notion of *disease stability* as a patient-specific latent trait could be a genuinely interesting contribution, particularly given your long-standing interest in variability as a meaningful biological signal rather than just statistical noise.

Here is your final analysis to make it a Master's research thesis:

Think of the paper in two parts.

Part I: Develop the hierarchical Bayesian progression model.

Part II: Explain *why* some patients have stable trajectories while others have unstable trajectories.

The quantity (\sigma_i) (patient-specific longitudinal variability) becomes a new phenotype that you try to explain.

What should predict stability?

Rather than throwing in every baseline variable from PPMI, I'd start from a biological hypothesis.

The literature increasingly suggests that **diffuse Lewy body pathology** produces a faster and more heterogeneous disease course than relatively focal nigrostriatal disease. Features associated with this diffuse phenotype repeatedly emerge as markers of worse progression.

I would therefore test whether baseline markers of diffuse disease predict **higher longitudinal variability** ((\sigma_i)).

The strongest candidates are:

Baseline feature	Why it is interesting
REM Sleep Behaviour Disorder (RBDSQ)	One of the strongest markers of diffuse α-synuclein pathology and consistently associated with faster progression. (PubMed)
Autonomic dysfunction (SCOPA-AUT)	Dysautonomia predicts more aggressive phenotypes and independently contributes to disease progression. (PubMed)
Olfaction (UPSIT)	Hyposmia is associated with more widespread pathology

Baseline feature	Why it is interesting
	and poorer longitudinal outcomes. (MDS Abstracts)
Cognition (MoCA)	Early cognitive impairment often accompanies a more diffuse disease process and faster progression. (PubMed)
DAT-SPECT deficit	Reflects severity of dopaminergic degeneration and may influence progression rate, although probably less strongly than the non-motor markers.
Age	Older onset is consistently associated with more rapid progression.

Notice something interesting.

These are **exactly** the variables clinicians already associate with a "malignant" PD phenotype.

The hierarchical model naturally extends

Instead of

$$\log(\sigma_i) \sim N(\mu_\sigma, \tau_\sigma^2)$$

make

$$\log(\sigma_i) = \gamma_0 + \gamma_1 RBD_i + \gamma_2 SCOPA_i + \gamma_3 UPSIT_i + \gamma_4 MoCA_i + \gamma_5 Age_i + u_i$$

Now you're saying

> Patient variability itself is explained by baseline biology.

This is much more interesting than simply using those variables to predict TD or PIGD.

Then your model estimates two things simultaneously

Progression

$$R_{it} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})t + \epsilon_{it}$$

Variability

$$\sigma_i = f(\text{baseline features})$$

The outputs become

- How fast does this patient progress?
- How predictable is that progression?

Those are different questions.

A conceptual interpretation

Imagine two patients.

Patient A

- Mild hyposmia
- No RBD
- Good cognition
- Low autonomic dysfunction

Model estimates

Progression slope = moderate, Variability = low

The model is confident.

Patient B

- RBD
- Severe hyposmia
- Mild cognitive impairment
- Autonomic dysfunction

Model estimates

Progression slope = similar, Variability = high

The average progression rate may even be the same, but the second patient's future is much less predictable. Clinically, that's valuable information because it indicates the patient should be monitored more closely.

I think there is an even stronger biological framing

Rather than calling (\sigma_i) **variability**, I'd call it something like:

Motor phenotype stability/ Trajectory stability/ Disease stability/ Progression stability index

Mathematically it is still the patient-specific residual standard deviation.

But biologically you hypothesize:

Patients with greater baseline evidence of widespread synucleinopathy (RBD, dysautonomia, hyposmia, cognitive dysfunction) exhibit **less stable motor phenotype trajectories**, leading to greater uncertainty in future subtype evolution.

That is a testable hypothesis, and it's a stronger story than saying "some patients have higher residual variance."

One suggestion that I think could make this paper distinctive

I would avoid using σ_i only as a nuisance parameter. Instead, elevate it to a **primary outcome**.

The paper could ask two parallel questions:

1. **What determines the speed of motor subtype progression?** (captured by the patient-specific slope b_{1i})
2. **What determines the stability of motor subtype progression?** (captured by the patient-specific variability σ_i)

Those two latent characteristics—**progression rate** and **progression stability**—are likely to reflect different aspects of Parkinson's disease biology. The first tells you *how quickly* a patient changes; the second tells you *how predictable* that change is. I think that distinction has the potential to make the work stand out both statistically and clinically.